# Vulnlog User Guide

## Table of Contents


1. What is Vulnlog? . . . 2
   1.1. Features . . . 2
   1.2. When to Use Vulnlog . . . 3
   1.3. How Vulnlog Compares . . . 3
      1.3.1. vs. Scanner-native suppression files . . . 3
      1.3.2. vs. Scanner vendor UIs . . . 3
      1.3.3. vs. Ticketing systems, spreadsheets, and Confluence pages . . . 4
   1.4. How Vulnlog Relates to Other Standards . . . 4
   1.5. Supported Reporters . . . 4
2. Getting Started . . . 4
   2.1. Creating a Vulnlog File . . . 4
   2.2. A Minimal Example . . . 5
   2.3. IDE Setup . . . 5
   2.4. CI Pipeline Integration . . . 5
3. Common Workflows . . . 6
   3.1. New finding — triage later . . . 6
   3.2. Adding the same finding to multiple branch files . . . 6
   3.3. Finding does not affect us . . . 7
   3.4. Not affected, but dependency updated anyway . . . 7
   3.5. Affected and fixed in the next release . . . 8
   3.6. Risk accepted with permanent suppression . . . 8
   3.7. Internal dependency, no customer impact . . . 8
4. Managing Multiple Release Branches . . . 9
5. CLI Reference . . . 9
   5.1. Usage . . . 9
   5.2. Filtering . . . 9
   5.3. Commands . . . 9
      5.3.1. `vulnlog init` . . . 9
      5.3.2. `vulnlog validate` . . . 10
      5.3.3. `vulnlog lint` . . . 10
      5.3.4. `vulnlog stats` . . . 10
      5.3.5. `vulnlog report` . . . 10
      5.3.6. `vulnlog suppress` . . . 10
      5.3.7. `vulnlog vex` . . . 10
      5.3.8. `vulnlog add` . . . 10
      5.3.9. `vulnlog copy` . . . 11

5.4. Exit Codes . . . 11
6. Full Example . . . 11

# 1. What is Vulnlog?

Vulnlog is a tool for tracking, analyzing, and managing software vulnerabilities reported by SCA (Software Composition Analysis) scanners.

It combines a simple YAML file format with a CLI that generates suppression files, HTML reports, and OpenVEX statements. The name combines "vulnerability" and "log" — Vulnlog is a persistent, version-controlled record of your team's vulnerability analysis.

For the full YAML format reference, see the Vulnlog Specification.

## 1.1. Features

**Vulnerability tracking**

Record and manage vulnerability findings from multiple SCA scanners in a single YAML file. Each finding captures the scanner report, affected packages, and the team's triage verdict.

**Verdict-driven workflow**

Classify vulnerabilities with a clear verdict: `affected`, `not_affected`, or `risk_acceptable`. Entries without a verdict are tracked as under investigation. Resolutions (dependency updates, patches) are recorded independently of the verdict.

**Flexible tagging**

Tag vulnerability entries with a team-defined taxonomy. Tags can represent deliverables (`binary`, `container`), usage contexts (`build-dep`, `test-infra`), or any other categorization.

**Suppression file generation**

Generate scanner-specific suppression files (`.trivyignore`, `.snyk`, etc.) from the Vulnlog file. Vulnerabilities with a resolution are excluded from suppression to verify the fix is complete.

**HTML report generation**

Generate self-contained HTML reports for communicating vulnerability status to team members, partners, and customers.

**OpenVEX statement generation**

Generate OpenVEX documents per release for standards-compliant vulnerability communication. Verdicts and resolutions map directly to OpenVEX statuses.

**Multi-scanner support**

Track findings from multiple SCA scanners in a single file. Switch scanners without losing analysis history.

**Non-scanner sources**

Record vulnerabilities from vendor advisories (e.g., Oracle CPU), penetration tests, and other

non-scanner sources alongside scanner findings.

**IDE integration**

JSON Schema published to SchemaStore provides validation, autocompletion, and inline documentation.

**Git-native**

Vulnlog files are plain YAML designed for version control. Changes are tracked via Git history. Each release branch has its own file, which archives naturally at end of life.

**Validation and linting**

Validate Vulnlog files against structural rules and cross-reference integrity. Lint for consistent field ordering and formatting.

**CI/CD integration**

The CLI runs on developer machines and in CI pipelines. Generate suppression files in CI to keep scanner pipelines green for triaged findings.

## 1.2. When to Use Vulnlog

Vulnlog is most valuable when your team:

- Uses multiple SCA scanners and needs a unified view of findings.
- Maintains multiple release branches and needs to track vulnerability decisions per branch.
- Needs to communicate vulnerability status via HTML reports or VEX statements.
- Wants to keep vulnerability analysis in the repository, next to the code.
- Wants to prevent stale suppression files that shadow real findings.
- Wants to avoid losing vulnerability analysis when moving to another SCA tool.

Vulnlog may be more than you need if you have a single scanner on a single branch — in that case, the scanner's native suppression mechanism may suffice.

## 1.3. How Vulnlog Compares

### 1.3.1. vs. Scanner-native suppression files

Scanner-native suppression files (`.trivyignore`, `.snyk`) track **what to suppress** but not **why**. Vulnlog generates these files from a richer decision record, keeping the **why** alongside the **what**.

### 1.3.2. vs. Scanner vendor UIs

Scanner vendor UIs (Snyk dashboard, GitHub Security tab, or Semgrep UI) create vendor lock-in: your triage decisions live in the vendor's platform. Vulnlog keeps your analysis in a YAML file you own.

### 1.3.3. vs. Ticketing systems, spreadsheets, and Confluence pages

Spreadsheets and wiki pages are disconnected from the code. Vulnlog lives in the repository and generates suppression files, reports, and VEX documents directly from the data.

## 1.4. How Vulnlog Relates to Other Standards

**VEX (Vulnerability Exploitability eXchange)**

Vulnlog generates OpenVEX documents. Vulnlog is the decision record; VEX is one of its outputs.

**SBOM (Software Bill of Materials)**

Complementary. SBOMs describe what components are in your software. Vulnlog tracks decisions about vulnerability findings in those components.

**SARIF (Static Analysis Results Interchange Format)**

A future Vulnlog output for integration with development tools.

## 1.5. Supported Reporters

| Type | Scanner / Source | Suppression Support |
|---|---|---|
| `trivy` | Trivy | Yes |
| `snyk` | Snyk | Yes |
| `dependency_check` | OWASP Dependency Check | Yes |
| `grype` | Grype | Planned |
| `rust_audit` | Cargo Audit (Rust) | Planned |
| `semgrep` | Semgrep | No |
| `npm_audit` | npm audit | No |
| `github_advisory` | GitHub Advisory Database | No |
| `other` | Generic / non-scanner source (e.g., Oracle CPU, penetration test reports) | No |

# 2. Getting Started

## 2.1. Creating a Vulnlog File

```
vulnlog init --project "My App" --organization "Acme Corp" --author "Security Team" -o
my-app.vl.yaml
```

**NOTE**

*File Name Convention*

For simple projects, `vulnlog.[yaml|yml]` are valid file names. For multiple Vulnlog

files, use `<project-name>.vl.[yaml|yml]`.

## 2.2. A Minimal Example

```
# $schema: ../generated/vulnlog-bundled.json
---
schemaVersion: "1"

project:
  organization: Acme Corp
  name: Acme Web App
  author: Acme Corp Security Team

releases:
  - id: 1.0.0
    published_at: 2026-01-15

vulnerabilities:

  - id: CVE-2026-1234
    releases: [ 1.0.0 ]
    description: Remote code execution in example-lib
    packages: [ "pkg:npm/example-lib@2.3.0" ]
    reports:
      - reporter: trivy
    analysis: >
      The vulnerable code path is not reachable in our application
      because we only use the safe subset of the API.
    verdict: not_affected
    justification: vulnerable_code_not_in_execute_path
```

## 2.3. IDE Setup

Vulnlog's JSON Schema is published to SchemaStore. IDEs with YAML support can automatically provide validation and autocompletion for `.vl.[yaml|yml]` and `vulnlog.[yaml|yml]` files.

To reference a local schema, add this as the first line of your Vulnlog file:

```
# $schema: .vulnlog/vulnlog-schema.json
```

## 2.4. CI Pipeline Integration

Example GitHub Actions workflow:

```
jobs:
  security-scan:
```

```
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Install Vulnlog CLI
        run: # install step

      - name: Generate Trivy suppression file
        run: vulnlog suppress my-app.vl.yaml --reporter trivy --release ${{
github.ref_name }}

      - name: Run Trivy scan
        uses: aquasecurity/trivy-action@master
        with:
          scan-type: fs
          trivyignores: .trivyignore

      - name: Generate vulnerability report
        if: always()
        run: vulnlog report my-app.vl.yaml --release ${{ github.ref_name }} -o
vulnlog-report.html

      - name: Upload report
        if: always()
        uses: actions/upload-artifact@v4
        with:
          name: vulnerability-report
          path: vulnlog-report.html
```

# 3. Common Workflows

## 3.1. New finding — triage later

```
  - id: CVE-2026-5678
    releases: [ 2.0.0 ]
    packages: [ "pkg:npm/image-lib@3.1.0" ]
    reports:
      - reporter: trivy
    tags: [ webapp ]
```

## 3.2. Adding the same finding to multiple branch files

Create stubs when no existing analysis exists yet:

```
vulnlog add branch-8.1.vl.yaml branch-8.2.vl.yaml \
```

```
  --vuln-id CVE-2026-5678 --reporter trivy
```

Or copy a complete entry when one branch already has the analysis:

```
vulnlog copy branch-8.3.vl.yaml branch-8.1.vl.yaml branch-8.2.vl.yaml \
  --vuln-id CVE-2026-5678
```

## 3.3. Finding does not affect us

```
  - id: CVE-2026-5678
    releases: [ 2.0.0 ]
    description: Memory corruption in image-lib
    packages: [ "pkg:npm/image-lib@3.1.0" ]
    reports:
      - reporter: trivy
        at: 2026-03-01
    tags: [ webapp ]
    analysis: >
      The vulnerable code path requires SVG input processing.
      Our application only accepts PNG and JPEG formats.
    analyzed_at: 2026-03-02
    verdict: not_affected
    justification: vulnerable_code_not_in_execute_path
```

## 3.4. Not affected, but dependency updated anyway

```
  - id: CVE-2026-5678
    releases: [ 2.0.0 ]
    description: Memory corruption in image-lib
    packages: [ "pkg:npm/image-lib@3.1.0" ]
    reports:
      - reporter: trivy
        at: 2026-03-01
    tags: [ webapp ]
    analysis: >
      The vulnerable code path requires SVG input processing.
      Our application only accepts PNG and JPEG formats.
    verdict: not_affected
    justification: vulnerable_code_not_in_execute_path
    resolution:
      in: 2.0.1
      note: "Updated image-lib from 3.1.0 to 3.2.0 during routine upgrade"
```

## 3.5. Affected and fixed in the next release

```
- id: CVE-2026-5678
  releases: [ 2.0.0 ]
  description: Memory corruption in image-lib
  packages: [ "pkg:npm/image-lib@3.1.0" ]
  reports:
    - reporter: trivy
      at: 2026-03-01
  tags: [ webapp ]
  analysis: >
    Confirmed exploitable via crafted PNG uploads.
  analyzed_at: 2026-03-02
  severity: high
  verdict: affected
  resolution:
    in: 2.0.1
    at: 2026-03-03
    ref: "https://jira.example.com/browse/SEC-99"
    note: "Updated image-lib from 3.1.0 to 3.2.0"
```

## 3.6. Risk accepted with permanent suppression

```
- id: CVE-2026-5678
  releases: [ 2.0.0 ]
  description: Memory corruption in image-lib
  packages: [ "pkg:npm/image-lib@3.1.0" ]
  reports:
    - reporter: trivy
      suppress: {}
  tags: [ webapp ]
  analysis: >
    Vulnerability confirmed but only exploitable with
    specially crafted input that our validation layer rejects.
  severity: medium
  verdict: risk_acceptable
  comment: >
    Risk accepted by project lead. Revisit when upstream
    releases a stable fix.
```

## 3.7. Internal dependency, no customer impact

```
- id: CVE-2026-9999
  releases: [ 2.0.0 ]
  description: Command injection in build-tool
  packages: [ "pkg:npm/build-tool@5.0.0" ]
```

```
    reports:
      - reporter: trivy
        at: 2026-03-01
    tags: [ build-dep ]
    analysis: >
      Affects the build pipeline only. The vulnerable command
      injection requires untrusted input which the CI environment
      does not expose.
    verdict: not_affected
    justification: vulnerable_code_cannot_be_controlled_by_adversary
```

# 4. Managing Multiple Release Branches

Maintain one Vulnlog file per release branch, all on the main branch:

```
my-app-8.1.vl.yaml    # Release branch 8.1.x
my-app-8.2.vl.yaml    # Release branch 8.2.x
my-app-8.3.vl.yaml    # Release branch 8.3.x
```

Use `vulnlog copy` to propagate complete analyses across multiple files.

# 5. CLI Reference

## 5.1. Usage

```
vulnlog <command> <file...> [flags]
```

Every command except `init` requires one or more Vulnlog files as positional arguments.

## 5.2. Filtering

| Flag | Description |
|---|---|
| `--release <id>` | Include vulnerabilities affecting all releases up to and including this release. |
| `--tags <id>` | Filter by tag (repeatable). |

## 5.3. Commands

### 5.3.1. `vulnlog init`

```
vulnlog init -o my-app.vl.yaml --project "My App" --organization "Acme Corp" --author
```

```
"Security Team"
```

### 5.3.2. vulnlog validate

```
vulnlog validate my-app.vl.yaml
vulnlog validate branch-8.1.vl.yaml branch-8.2.vl.yaml --strict
```

### 5.3.3. vulnlog lint

```
vulnlog lint my-app.vl.yaml          # Check only
vulnlog lint my-app.vl.yaml --fix    # Rewrite with corrected ordering
```

### 5.3.4. vulnlog stats

```
vulnlog stats my-app.vl.yaml
vulnlog stats branch-8.1.vl.yaml branch-8.2.vl.yaml branch-8.3.vl.yaml
vulnlog stats my-app.vl.yaml --release 8.1.1
```

### 5.3.5. vulnlog report

```
vulnlog report my-app.vl.yaml
vulnlog report my-app.vl.yaml --release 8.1.1 -o report.html
```

### 5.3.6. vulnlog suppress

```
vulnlog suppress my-app.vl.yaml --reporter trivy --release 8.1.1
vulnlog suppress my-app.vl.yaml
vulnlog suppress my-app.vl.yaml --reporter trivy -o custom/.trivyignore
```

### 5.3.7. vulnlog vex

```
vulnlog vex my-app.vl.yaml --release 8.1.1
vulnlog vex my-app.vl.yaml --release 8.1.1 --tags binary -o vex-binary.json
```

### 5.3.8. vulnlog add

```
# Create stub entries in multiple files
vulnlog add branch-8.1.vl.yaml branch-8.2.vl.yaml \
  --vuln-id CVE-2026-001 --reporter trivy
```

### 5.3.9. `vulnlog copy`

```
# Copy a complete entry from the main branch file to older branch files
vulnlog copy branch-8.3.vl.yaml branch-8.1.vl.yaml branch-8.2.vl.yaml \
  --vuln-id CVE-2026-001
```

## 5.4. Exit Codes

| Code | Name | Description |
|---|---|---|
| `0` | Success | Command completed successfully. |
| `1` | General error | Unexpected error. |
| `2` | Validation error | The Vulnlog file contains validation errors. |
| `3` | File not found | The specified file does not exist. |
| `4` | Lint violation | The file has lint violations (without `--fix`). |
| `5` | Missing required flag | A required flag was not provided. |
| `6` | Invalid flag value | A flag references an unknown entity or unsupported type. |

# 6. Full Example

```
# $schema: ../generated/vulnlog-bundled.json
---
schemaVersion: "1"

project:
  organization: Demo Project Inc
  name: Demo Project
  author: Demo Security Team
  contact: security@demo.com

tags:
  - id: binary
    description: Demo Project binary file
  - id: container
    description: Demo Project container image
  - id: sdk
    description: Demo Project SDK
  - id: build-dep
    description: Internal build dependency

releases:
  - id: 8.1.0
    published_at: 2021-01-15
```

```
    purls:
      - purl: "pkg:generic/demo-project@8.1.0"
        tags: [ binary ]
      - purl: "pkg:docker/demo-project@8.1.0"
        tags: [ container ]
  - id: 8.1.1
    published_at: 2021-02-11
    purls:
      - purl: "pkg:generic/demo-project@8.1.1"
        tags: [ binary ]
      - purl: "pkg:docker/demo-project@8.1.1"
        tags: [ container ]
  - id: 8.1.2

vulnerabilities:

  # Report from a non-scanner source (Oracle CPU)
  - id: CVE-2022-21449
    name: Psychic Signatures
    releases: [ 8.1.2 ]
    description: ECDSA signature verification bypass in Java SE
    packages: [ "pkg:maven/com.oracle/jdk@17.0.2" ]
    reports:
      - reporter: other
        at: 2022-04-19
        source: "Oracle Critical Patch Update April 2022"
    tags: [ binary, container ]
    analysis: The project uses Java 17 and ECDSA signatures for JWT token
verification. Confirmed exploitable.
    analyzed_at: 2022-04-20
    severity: critical
    verdict: affected
    resolution:
      in: 8.1.2
      at: 2022-04-22
      ref: "https://jira.example.com/browse/SEC-55"
      note: "Updated JDK from 17.0.2 to 17.0.3"

  # Verdict: risk_acceptable - vulnerability present but risk accepted
  - id: SNYK-2021-003
    releases: [ 8.1.2 ]
    description: DoS vulnerability in some-other-dependency
    packages: [ "pkg:maven/com.some.other/some-other-dependency@3.11.0" ]
    reports:
      - reporter: snyk
        at: 2022-04-28
    tags: [ binary, container ]
    analysis: >
      Demo project utilizes some-other-dependency but only in rare cases.
      Therefore the risk is considered low and a fix would potentially
      be breaking for customer setups.
```

```yaml
    analyzed_at: 2022-04-29
    comment: >
      Risk accepted by project lead (decision documented in RISK-12).
      Revisit when some-other-dependency 4.0 is released.
    severity: low
    verdict: risk_acceptable

  # Verdict: not_affected - OS-level vulnerability in container base image
  - id: CVE-2024-5535
    releases: [ 8.1.1 ]
    description: Buffer overread in OpenSSL SSL_select_next_proto
    packages: [ "pkg:apk/alpine/openssl@3.1.4-r0" ]
    reports:
      - reporter: trivy
        at: 2024-06-28
    tags: [ container ]
    analysis: >
      Vulnerable OpenSSL is present in the container base image but the affected
      NPN/ALPN negotiation code path is never invoked by our application.
    analyzed_at: 2024-06-29
    verdict: not_affected
    justification: vulnerable_code_not_in_execute_path

  # Verdict: affected - resolved in a subsequent release (named vulnerability)
  - id: CVE-2021-44228
    name: Log4Shell
    aliases: [ GHSA-jfh8-c2jp-5v3q ]
    releases: [ 8.1.1 ]
    description: RCE vulnerability in Log4J via JNDI lookup
    packages: [ "pkg:maven/org.apache.logging.log4j/log4j-core@2.14.1" ]
    reports:
      - reporter: trivy
        at: 2021-12-10
    tags: [ binary ]
    analysis: >
      Demo project utilizes Log4J and a PoC showed that the project
      is vulnerable to this RCE vulnerability.
    analyzed_at: 2021-12-10
    comment: >
      PoC exploit confirmed by Jane (security team) on 2021-12-10.
      See internal report attached to SEC-42.
    severity: critical
    verdict: affected
    resolution:
      in: 8.1.2
      at: 2021-12-12
      ref: "https://jira.example.com/browse/SEC-42"
      note: "Updated Log4J from 2.14.1 to 2.17.1"

  # Verdict: not_affected - with defensive resolution
  - id: CVE-2021-002
```

```
    aliases: [ GHSA-xxxx-yyyy ]
    releases: [ 8.1.1 ]
    description: Deserialization vulnerability in some-dependency
    packages: [ "pkg:maven/example.org/some-dependency@1.1.2" ]
    reports:
      - reporter: trivy
        at: 2021-03-03
      - reporter: snyk
        at: 2021-03-03
        vuln_id: [ SNYK-2021-002 ]
    tags: [ binary, container ]
    analysis: >
      Demo project does not utilize some-dependency
      and therefore is not affected.
    analyzed_at: 2021-03-03
    verdict: not_affected
    justification: vulnerable_code_not_present
    resolution:
      in: 8.1.2
      note: "Removed unused dependency some-dependency"

  # Stub entry - quick triage, no description, no verdict yet
  - id: CVE-2021-007
    releases: [ 8.1.1 ]
    packages: [ "pkg:maven/com.example/some-lib@4.0.0" ]
    reports:
      - reporter: trivy
        at: 2021-03-10
    tags: [ binary ]

  # No verdict - under investigation, with temporary suppression
  - id: CVE-2020-099
    releases: [ 8.1.0 ]
    description: Path traversal vulnerability in random-dependency
    packages: [ "pkg:maven/org.random/random-core@1.33.7" ]
    reports:
      - reporter: trivy
        at: 2020-12-28
        suppress:
          expires_at: 2021-01-02
    tags: [ build-dep ]
    analysis: >
      Vulnerability in internal build dependency.
      No fixed library available at the moment. Re-check later.
    comment: >
      Checked upstream issue tracker - fix expected in random-core 1.33.8.
      Monitor https://github.com/random/random-core/issues/421
    severity: low
```